

Ali Shan

AI & Machine Learning Engineer | Data Scientist | Python Developer

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PROFILE

Results-driven AI Engineer and Data Scientist with 2+ years of experience building and shipping end-to-end machine learning systems across computer vision, NLP, generative AI, and data analytics. Experienced in taking projects from research and data preparation through production deployment, dashboards, automation, and cloud/edge delivery. Strong background in Python, machine learning, full-stack integration, and translating technical requirements into practical solutions.

CORE TECHNICAL SKILLS

Languages	Python, C++, Java, JavaScript/TypeScript, Dart, Bash
ML / DL	PyTorch, TensorFlow/Keras, HuggingFace Transformers, OpenCV, YOLOv5/8
Data Science	Pandas, NumPy, Statistical Analysis, Feature Engineering, Predictive Modeling
GenAI & NLP	LangChain, RAG Pipelines, Prompt Engineering, Diffusion Models, LLM Fine-tuning
Visualization	Power BI, Tableau, Matplotlib
Full-Stack & Serving	FastAPI, Flask, React/Next.js, Node.js, Streamlit, Gradio
MLOps & Cloud	Docker, GitHub Actions, AWS S3/EC2, GCP Vertex AI, ONNX, NVIDIA TensorRT
Databases	MongoDB, PostgreSQL, Firebase Firestore

PROFESSIONAL EXPERIENCE

Data Scientist | 04 Nov 2025 - Present

Chang Sheng Agro Chemical Pvt Ltd — Lahore, Pakistan

- Develop and deploy data-driven solutions to optimize agricultural chemical production and supply-chain processes, including predictive models for demand forecasting, crop-yield analysis, and disease-pattern prediction.
- Perform data analysis and preprocessing using Python (Pandas, NumPy), applying statistical techniques and feature engineering to improve model performance and reliability.
- Create interactive dashboards and analytical visualizations using Power BI, Tableau, and Matplotlib to communicate operational insights to stakeholders.
- Automate data pipelines and reporting workflows to improve operational efficiency, reduce manual reporting effort, and support timely decision-making.
- Collaborate with cross-functional teams to translate business and operational requirements into practical data science solutions.

AI Engineer (Freelance / Contract) | Jan 2023 - Present

Independent Consultant — Remote

- Delivered 8+ end-to-end client projects spanning healthcare, security, and e-commerce, owning pipelines from data collection through deployment.
- Built production ML pipelines with PyTorch, served through FastAPI, and integrated models into React and Flutter applications.
- Applied pruning and quantization to computer-vision models, achieving up to 3x faster inference with negligible accuracy loss.
- Established CI/CD workflows using GitHub Actions with Dockerized, automated deployments.

Artificial Intelligence Intern | Jul 2025 - Oct 2025

AI SoftDevs — Remote

- Built an AI content-to-PDF enhancer in React that converts generated text into customizable, share-ready documents.
- Developed a Prompt-to-JSON Chrome extension that structures natural-language prompts into machine-readable schemas.
- Designed an AI stock-market analysis agent and an automated blogging agent for scheduled multi-platform publishing.
- Contributed to full-stack development, workflow optimization, technical documentation, and live client demos.

Machine Learning Intern | Jun 2024 - Aug 2024

Pakistan Ordnance Factories (POF) — Wah Cantt, Pakistan

- Prototyped a soldier-worn thermal-vision concealed-weapon detector, reducing false positives by 18% versus the YOLOv5s baseline.
- Optimized model performance for constrained edge hardware through data preprocessing and inference tuning.
- Authored an internal whitepaper on edge-deployment strategies for defense-grade computer vision.

EDUCATION & TRAINING

B.Sc. (Honours) in Computer Science | 2021 - 2025

COMSATS University Islamabad, Wah Campus

CGPA: 3.17 / 4.00

FCS (Computer Science) — A+ Grade | 2019 - 2021

F.G. Degree College for Men, Wah Cantt

A+

Matriculation (SSC) — A+ | EQF Level 3 | 2017 - 2019

F.G. Public School No. 3, Wah Cantt, Pakistan

A+

SELECTED PROJECTS

AI Content-to-PDF Enhancer — React-based live application that converts generated content into customizable, share-ready PDF documents. [\[Live Project\]](#)

RF-DETR for Concealed Weapon Detection — Region-Focused DETR built in PyTorch to localize firearms in thermal imagery; achieved mAP 0.71 and integrated into a live CCTV pipeline.

MedGPT — Domain RAG Chatbot — Retrieval-augmented assistant grounding a fine-tuned Llama-2 model on BM25-retrieved PubMed abstracts; deployed with LangChain and Streamlit.

AnimeGAN Creator — Full-stack Flutter application transforming selfies into anime-style art using a custom-trained AnimeGAN-v3 model, served through FastAPI and ONNX Runtime on AWS.

Skin Cancer Detector — ResNet-50 fine-tuned on ISIC 2019 with 92% balanced accuracy, quantized with INT8/TFLite for real-time on-device inference.

Metal Implant Detection (X-Ray) — YOLOv8-nano model flagging orthopedic implants with sub-40ms latency on NVIDIA Jetson Nano.

ADDITIONAL PROJECTS

Emotion Detector (NLP) • Satellite House Detection • Gender Classification via Eye • Cat-Dog Classifier • Heart Disease Risk Predictor • Flappy-Bird Clone

CERTIFICATIONS & LEADERSHIP

Kaggle — Machine Learning, Deep Learning & Python • Walmart — Advanced Software Engineering Job Simulation • HP LIFE — Data Science & Analytics • Codanics — Data Science Bootcamp

Lead ML Mentor, Google Developer Student Clubs — organized 6 workshops on Computer Vision & Generative AI
Winner, COMSATS AI Hackathon 2024 — real-time flood-detection drone vision system (top team of 30)

LANGUAGES & INTERESTS

Urdu — Native • English — Professional (C1 written) • Chinese — Basic (A1)

Open-source contribution (YOLO, FastAPI) • AI Ethics • Sci-Fi Literature • Badminton

REFERENCES

Dr. Muhammad Khalil Afzal

HOD | COMSATS University Islamabad, Wah Campus

I have known him for three years as my teacher during my 3rd semester at COMSATS and as one of my strongest recommenders.

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Dr. Saeed Ur Rehman

Tenured Associate Professor | COMSATS University Islamabad

I have known him for one year; he was one of my teachers when I was at COMSATS and is one of my strongest recommenders.

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